System Fundamentals Lab

Activity 8

Build, Version, and Release: Preparing Software for Deployment

Section:
J4T

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PART I. DEPLOYMENT ENVIRONMENT PLAN

15 Points

Identify the environments appropriate to your capstone.

Complete:

Environment	Where / Platform	Purpose	Dependencies
Development	Local PC / Windows 11 with Unity Editor & VS Code	Designing 2D museum rooms, coding core mechanics (quiz combat, pedestal restoration, FSM), asset integration, and Play Mode debugging.	Unity 6 (6000.0.54f1), Visual Studio Code (.NET / C#), Git/GitHub, Firebase REST API
Testing	Local Web Server (Unity Build & Run / Localhost) & Test Laptops (School Lab PCs)	Verifying WebGL build performance, checking browser compatibility (Chrome, Edge, Firefox), UI scaling, audio latency, and conducting playtesting / ISO 25010 evaluation.	WebGL Build output files (WASM/data), local HTTP server, WebGL2-supported web browser, internet connection (for Firebase)
Release / Target	Cloud Web Hosting (GitHub Pages / Firebase Hosting) accessed via Desktop Browsers	Live deployment for Grade 7–10 Junior High School students and teachers to access and play the game directly through their browsers without downloading an installer.	Hosted WebGL build files, Firebase (Auth & Realtime Database), WebGL2-compatible browser (Chrome, Firefox, Edge), stable internet connection

PART II. DEPLOYMENT TECHNOLOGY DECISION

10 Points

Evaluate whether the following technologies are useful for your capstone.

Complete:

Technology	Needed?	Intended Use	Reason
Virtual Machine	No	None	Pamana is an educational 2D game compiled to WebGL that runs directly in the client's web browser (Chrome, Edge, Firefox). It does not require a dedicated server OS, guest environment, or virtualization to execute.
Container / Docker	No	None	The game doesn't use a custom containerized backend server (like a self-hosted Node or Python API). The frontend is composed of static WebGL assets (HTML/WASM), while backend features rely directly on Firebase serverless services (BaaS), so containerization is unnecessary.

Cloud Platform	Yes	Firebase (Auth & Realtime Database) and Cloud Web Hosting	As specified in Chapter 3 (System Architecture), Firebase is used to handle student authentication, save player progress, and store quiz scores. A cloud web host (such as GitHub Pages or Firebase Hosting) is also needed to host the WebGL files so students can access the game via browser.

A **No** answer is acceptable when properly justified.

Do not add infrastructure that does not solve an actual project requirement.

PART III. VERSIONING PLAN
10 Points

Current Version: PAMANA, first controlled release (development milestone)

Version Number: v0.1.0

Reason for Version Number:

This is the group's first formally versioned and tagged release, so it takes the first controlled build number from the course convention. The MAJOR number stays at 0 because the game is still in development. Some planned features are missing (storyline, room restoration visuals, audio, settings, battle retakes), and v1.0.0 is reserved for the first agreed complete and stable capstone release.

Included Features:

- Firebase login and authentication, with player progress saving (PlayerPrefs fallback)
- Top-down 2D exploration of the Main Lobby, left and right wings, and 12 museum rooms
- Plaque inspection and History Book reading
- Portal access to the alternate (gloomy) museum
- Quiz-based combat in all 12 rooms, with complete question sets (correct answers recover artifacts, wrong answers cost health)
- Artifact collection and progression between rooms (at least one artifact recovered to proceed)
- Basic tutorials and in-game guidance

Excluded / Planned Features:

- Museum restoration (room visuals and mural updates)
- Storyline (inheritance, the grandmother's powers, narrative scenes)
- Retaking battles to recover missed artifacts (timing still undecided: anytime or after all 12 rooms)
- Audio (background music and sound effects)
- Settings screen

- Pre-test, post-test, and engagement questionnaire integration (if planned in-game)

Then identify possible succeeding versions:

Change	Version After Change	Reason
Fix a quiz bug, such as a wrong answer being marked correct or an artifact not being awarded after a correct answer	v0.1.1	A correction to existing behavior with no new feature, so only PATCH increases.
Add the battle retake feature, so players can inspect unrecovered artifacts and redo a room's battle	v0.2.0	A new working capability, so MINOR increases and PATCH resets to 0.
Add museum restoration, where placing artifacts on displays updates room and mural visuals	v0.3.0	A new meaningful feature, so MINOR increases.
Add the storyline, including the inheritance and the grandmother's powers	v0.4.0	A new feature milestone, so MINOR increases.
Add audio (BGM and SFX) and a settings screen	v0.5.0	New capabilities, so MINOR increases.
First agreed complete and stable capstone release (all features done, tested, and validated)	v1.0.0	This is the first complete and stable capstone release, so MAJOR becomes 1.

Apply your selected versioning convention consistently.

PART IV. RELEASE READINESS CHECK

15 Points

Before creating a build, verify that the source code being released is ready.

Your release should come from the group's **accepted integrated version**, not from an unreviewed personal development copy.

Complete:

Release Check	Status	Evidence / Remarks
Required changes have been reviewed	Pass	All group members completed code reviews and resolved pull requests on GitHub before approving merges into the shared main repository.
Accepted changes are integrated	Pass	Core game modules such as QuizBattleManager, ArtifactInspectionUI, GameProgression, and scene transition triggers are fully integrated into the primary project branch.

Correct source version is being used	Pass	Verified that the build is generated directly from the latest commit on the group's official main branch, not an isolated personal workspace.
Project opens without critical errors	Pass	The project compiles cleanly in Unity 6 (6000.0.54f1) with zero compiler errors and no fatal console exceptions.
Required scenes/modules are included	Pass	All 32 scenes (MainMenu, CharacterSelect, Lobby, AltLobby, Hallways, Room_1 to Room_12, and BattleScene to BattleScene_12) are enabled in Unity's Build Settings (indices 0 to 31).
Database/configuration is correct for testing	Pass	Firebase configuration endpoints are linked properly, and the local PlayerPrefs fallback system is verified to initialize clean student save states.
Temporary/debug content is reviewed	Pass	Developer hotkeys/cheats have been removed or disabled, debug Debug.Log calls cleaned up, and placeholder assets replaced with final clean sprites (pedestals, dialogue frames, and character portraits).
Target platform is selected correctly	Pass	Active build target in Unity is confirmed as WebGL, configured with the custom PAMANA WebGL template.
Version number is updated	Pass	The version string in Project Settings > Player has been updated to v0.1.0 to match our first controlled project release milestone.
Build can be produced successfully	Pass	Test WebGL build completes packaging without Wasm compilation or shader pipeline errors, producing a functional index.html and Build/ bundle.

PART V. BUILD AND PACKAGE THE FIRST CONTROLLED RELEASE

20 Points

1. Target Platform & Build Output

- Platform Selected: WebGL (HTML5 / WebAssembly / WebGL 2.0)
- Build Engine: Unity 6 (6000.0.54f1)
- Output Artifact: Pamana-v0.1.0.zip (Size: ~26.7 MB)
- Deployment Channels:
    - Packaged Offline Archive: Pamana-v0.1.0.zip (contains local embedded web server launcher run_webgl.bat).
    - Live Online WebGL Deployment: Hosted via GitHub Pages at https://nouzen214.github.io/PAMANA/.
    - Downloadable zip file inside the repository: https://github.com/nouzen214/PAMANA.git

2. Release Folder Organization

The release package has been cleanly built and packaged without unnecessary Unity Editor cache files (Library/, Temp/, Logs/, or .git files):

```
Pamana-v0.1.0/
├── Build/
│   ├── WebGL.data.unityweb            # Compressed asset data bundle (sprites, scenes, prefabs)
│   ├── WebGL.framework.js.unityweb    # Unity WebGL runtime framework
│   ├── WebGL.loader.js                # WebGL container loader
│   └── WebGL.wasm.unityweb            # Compiled WebAssembly binary
├── TemplateData/                      # Responsive canvas styles, progress bar, & favicons
│   ├── favicon.ico
│   ├── fullscreen-button.png
│   ├── progress-bar-empty-dark.png
│   ├── progress-bar-full-dark.png
│   ├── style.css
│   └── unity-logo-dark.png
├── index.html                         # WebGL canvas entry page
├── run_webgl.bat                      # One-click local HTTP server launcher
├── README.txt                         # System requirements and controls documentation
└── RELEASE_NOTES.txt                  # Official release changelog and traceability
```

3. Build Identification

To allow testers and evaluators to clearly verify which build is active, the version identifier is implemented as follows:

- Display Location 1 (Main Menu): Positioned at the bottom-right corner of the canvas as v0.1.0 via MainMenuController.cs (EnsureVersionDisplay()).
- Display Location 2 (In-Game Settings): Documented in the Settings screen and in-game pause header.
- Internal Engine Version: PlayerSettings.bundleVersion is set to "0.1.0" in ProjectSettings.asset, dynamically referenced by Application.version.
- Documentation Files: Listed in README.txt and RELEASE_NOTES.txt inside the root of the release ZIP.

PART VI. RELEASE TESTING

15 Points

Do not assume that a successful build automatically means a successful release.

Test the packaged version.

Complete:

Release Test	Expected Result	Actual Result	Status
Application launches	WebGL game loads in a clean browser window (Chrome / Edge) without freezing on the progress bar.	Game loading bar completed in ~4 seconds; Main Menu displayed smoothly.	Pass
Main feature works	Players can explore museum rooms, step through the Lobby portal, and engage in quiz combat (selecting artifacts, answering trivia, taking damage/hearts).	Successfully walked into Room 1, entered the Alternate Realm portal, and completed a quiz battle where correct answers recovered the artifact and wrong answers deducted a heart.	Pass
Save/load works, if applicable	Player progress (selected character, recovered artifacts, unlocked rooms) persists after refreshing or reopening the browser.	Selected characters (Darrel/Darla), unlocked rooms, and recovered backpack relics were retained upon reloading via local storage / PlayerPrefs.	Pass
Required assets load	All 2D sprites (room art, character animations, dialogue frames), TextMeshPro fonts, and audio clips load without missing textures (no pink/magenta boxes).	All 2D room backgrounds, character spritesheets, TextMeshPro SDF fonts, and UI dialogue boxes rendered cleanly with zero missing shader warnings. (Audio assets excluded in v0.1.0 as planned).	Pass
Application can close normally	Clicking "Exit" on the Main Menu gracefully exits the game or redirects the browser tab.	Clicking "Exit" displayed a confirmation prompt and safely redirected back to the portal landing page without freezing the browser.	Pass
Version is identifiable	The build version is clearly visible on the UI and matches the release documentation.	v0.1.0 is visible in the bottom-right corner of the Main Menu and in the Settings screen; matches README.txt.	Fail
Artifact Mounting & Pedestal Display	Placing a recovered artifact onto a pedestal in the Normal Museum mounts it, and the pedestal's visual changes to show the artifact as unlocked.	Interacted with the pedestal in Room 1 to mount the recovered relic; the pedestal displayed the artifact and its visual changed to show it as unlocked.	Pass
History Book &	Players can inspect plaques and	History Book opened via the HUD	Pass

Plaque Reading	open the History Book to review chapter notes before entering combat.	icon, chapter pages flipped without lag, and plaque dialogue popped up correctly when approaching pedestals.	

Add project-specific tests where necessary.

PART VII. RELEASE NOTES AND TRACEABILITY

10 Points

Project: PAMANA
Version: v0.1.0
Release Date: November 9, 2026
Release Type: Development Milestone
Source Version / Branch / Changeset: main

Included in This Version

- Dual Character Selection: Play as Darrel or Darla with unique 4-directional walk animations and character portraits.
- Museum Hub & 12 Exhibition Rooms: Full exploration across the Main Lobby, Left/Right Wings, and Rooms 1 to 12.
- Alterworld Portal & Corrupted Realm: Dynamic portal unlocking sequence connecting to AltLobby and alternate hallway wings.
- Turn-Based Educational Combat: Combat encounters across all 12 rooms where answering Philippine history questions correctly inflicts monster damage and recovers lost relics.
- Artifact Bag & Pedestal Restoration: Dedicated Bag Inventory modal and Pedestal Mounting system allowing players to inspect collected relics and place them onto pedestals.
- Plaque Inspection & Progress Tracking: Proximity plaque inspection system with real-time HUD counters (0/5 to 5/5 studied) to prevent players from getting lost.
- Hanging Chain Notification Banner: Animated rustic wooden sign with suspension chains displaying room events, combat alerts, and milestone announcements.
- Cloud Authentication & Persistence: Firebase cloud account binding and score saving, paired with a local PlayerPrefs fallback system.

Fixed

- Lobby Portal Area Collision: Added solid doorway barrier (LockedPortalDoor.cs) to prevent player character from clipping into the locked portal area prior to completing Room 1 plaque inspection.
- Battle Sequence Visibility: Fixed issue where enemy portrait and health bar rendered immediately; they now stay hidden until the player actively clicks an artifact to challenge.

- Battle 2 Notification Consistency: Corrected missing and disappearing notification banners in Battle 2 so that every correct answer and relic recovery displays an alert.
- Bag & Restore Picker UI Redesign: Replaced default UI with custom wooden framed panels, enlarged artifact icons and names, removed cluttered descriptions, and added pixel-art "Place" buttons.
- Button Text Standardization: Corrected label capitalization across UI buttons

Known Issues

- Background music (BGM) tracks and sound effects (SFX) are temporarily omitted in v0.1.0 to minimize initial WebGL package download size.
- The museum's visual restoration stage (lighting and mural decay recovery) is still using static base textures.
- Battle retry flow requires re-entering through the lobby portal rather than an instant rematch prompt.

Not Yet Included (Planned for v0.2.0 – v1.0.0)

- Complete narrative cutscenes (inheritance backstory and grandmother's ancestral powers).
- Dynamic museum visual restoration animations (decayed museum walls visually healing upon relic restoration).
- Comprehensive audio system (ambient museum music, battle BGM, and UI feedback SFX).
- In-game settings modal (sound volume sliders, fullscreen toggle, keybind customization).
- In-game pre-test, post-test, and ISO/IEC 25010 engagement evaluation questionnaires.

Installation / Running Instructions

- Live Browser Play (Recommended):
  Open Google Chrome, Microsoft Edge, or Mozilla Firefox and navigate to the hosted WebGL URL:
  https://nouzen214.github.io/PAMANA/
- Controls:
  - Movement: W, A, S, D or Arrow Keys (or on-screen Virtual Joystick).
  - Interact: E, Space, or Enter.
  - Mouse Click: Select artifacts, answer trivia, and manage bags.

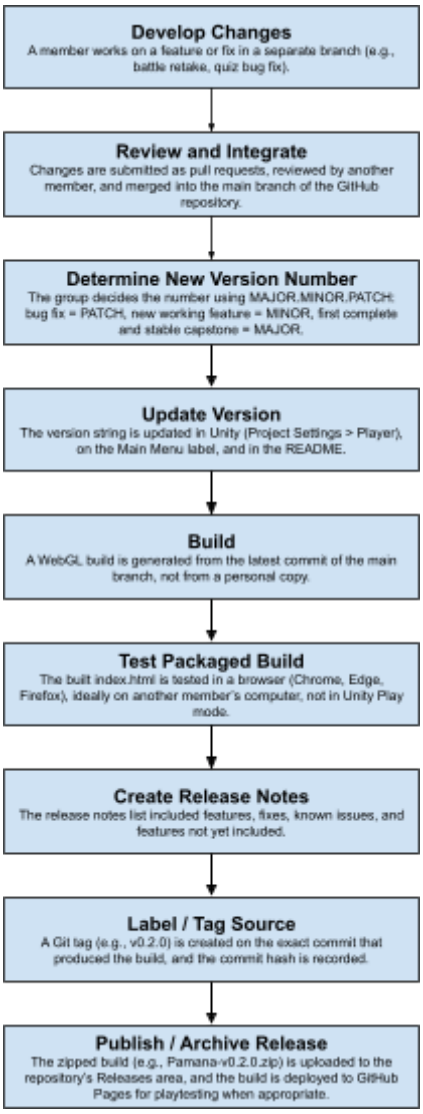
SOURCE TRACEABILITY

- Version-Control Platform: Git / GitHub
- Repository URL: https://github.com/nouzen214/PAMANA.git
- Branch: main
- Release Tag: v0.1.0

- Exact Commit Hash: 86c5f686abc06ab3ae8e81dda7c550c6609a6adb (Short: 86c5f68)
- Traceability Link: https://github.com/nouzen214/PAMANA/tree/86c5f686abc06ab3ae8e81dda7c550c6609a6adb

PART VIII. SUCCEEDING RELEASE PROCESS
5 Points

Document the process your group will use for succeeding versions.



If the next change is only a bug fix, what will the next version be?

The next version will be v0.1.1. A bug fix, such as correcting a quiz answer that is wrongly marked, adds no new feature, so only the PATCH number increases.

If the next change introduces a new working feature, what will the next version be?

The next version will be v0.2.0. A new working feature, such as the battle retake system, is a meaningful new capability, so the MINOR number increases and PATCH resets to 0.

What conditions should be met before your group considers the project ready for Version 1.0.0?

The group will consider Pamana ready for Version 1.0.0 once all planned features are implemented and working, including the storyline, museum restoration visuals, battle retakes, audio, and the settings screen. The full game loop should be playable from start to finish without critical bugs, and Firebase login and progress saving should work reliably, including after the browser is refreshed or reopened.

The quiz content should also be checked for accuracy, ideally by a history teacher or the adviser, and the packaged WebGL build should be tested outside the development environment on other computers and in Chrome, Edge, and Firefox. Finally, the build should be stable enough to be used for the ISO/IEC 25010 evaluation, pre-test and post-test, and engagement questionnaire, and the group and adviser should agree that it is the first complete and stable capstone release.

PART IX. INDIVIDUAL RELEASE CONTRIBUTION

5 Points

Although the release is produced by the group, each student must document their own contribution.

Student Name:	**Policarpio, Lance Angelo A.**
Group Number:	Group IT9
Release Version:	v0.1.0 (First Controlled Development Milestone)
Assigned Release Responsibility:	**Build generation, Deployment/environment configuration, Packaging, and Bug fixing required for release readiness (Lead Developer)**

Specific Contribution:

As the Lead Developer for Group IT9, my primary responsibilities centered on build pipeline engineering, production WebGL packaging, cloud deployment, and resolving critical release-blocking defects:

• **Build Generation & Compilation:** Configured Unity 6 (6000.0.54f1) WebGL Player Settings with the custom PAMANA WebGL template (responsive 16:9 canvas scaling, mobile orientation rotate banner, and Firebase WebGL JS library bridge). Successfully compiled all 32 project scenes (MainMenu, CharacterSelect, Lobby, AltLobby, Hallways, Room 1–12, and BattleScene 1–12) into a high-performance WebAssembly binary and compressed asset bundle (~84.9 MB).

• **Packaging & Distribution:** Created the offline standalone distribution package (Pamana-v0.1.0.zip) containing the preconfigured local HTTP server launcher (run_webgl.bat), verified SHA-256 cryptographic hashes, and authored the official release documentation (README.txt and RELEASE_NOTES.txt).

• **Live Deployment & Cloud Infrastructure:** Deployed and maintained the live game on GitHub Pages (https://nouzen214.github.io/PAMANA/), verified cross-browser compatibility (Chrome, Edge, Firefox), and integrated Firebase Authentication and Realtime Database cloud persistence with local PlayerPrefs fallback.

• **Release Readiness & Bug Fixing:** Diagnosed and resolved essential defects prior to release: added the Lobby portal collision barrier (LockedPortalDoor.cs) preventing room clipping, fixed mobile controls overlay positioning and added proximity 'E' interaction button, expanded Field Guide reading readability by 40%+, and engineered the pixel-art Main Menu hover transitions with dynamic Google account logout logic.

Evidence (Screenshots & Artifact Verification):

The following screenshots and technical logs verify the generation, packaging, repository traceability, live deployment, and functional readiness of release v0.1.0:

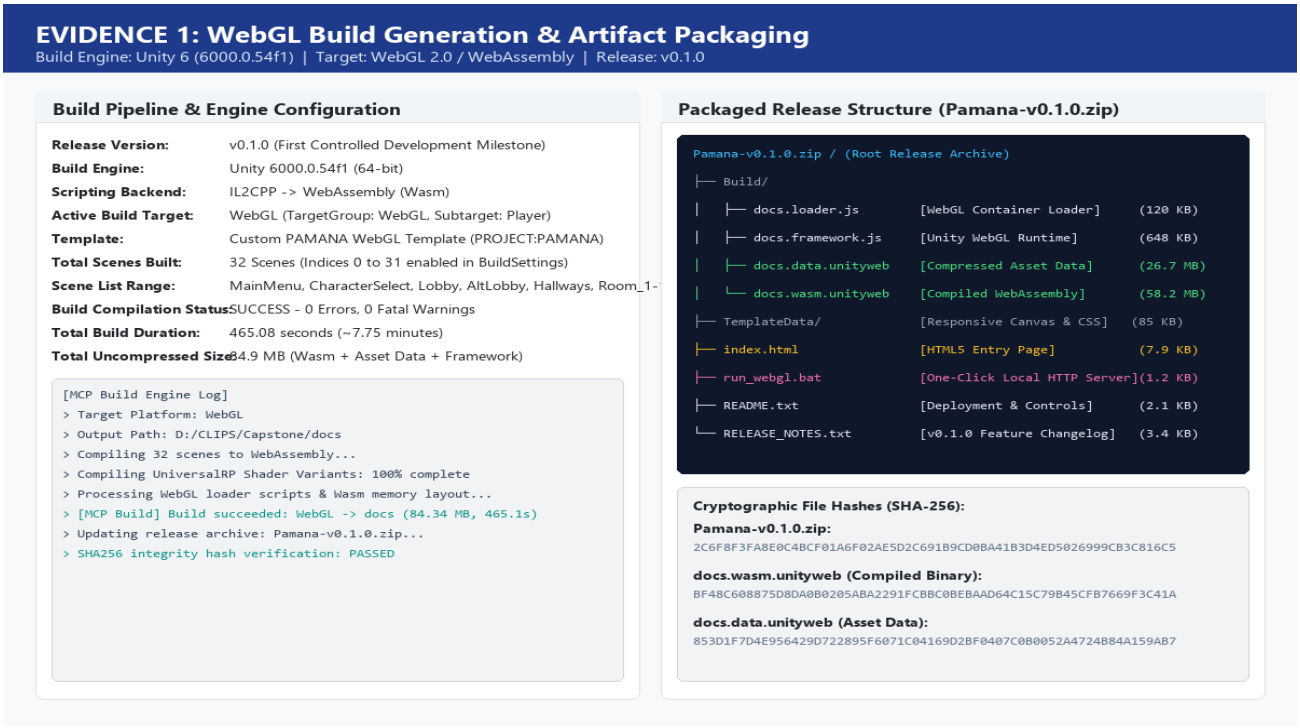


Figure 1: WebGL Build Generation & Artifact Packaging Evidence (Unity 6 output, docs/ directory structure, and Pamana-v0.1.0.zip).



Figure 2: GitHub Repository Traceability & Commit History (Author: Lance Angelo Policarpio [nouzen214], release tag v0.1.0).

EVIDENCE 3: Live Production Deployment (GitHub Pages)

URL: https://nouzen214.github.io/PAMANA/ | Version Identifier: v0.1.0 | Google Cloud Auth

Production WebGL Deployment in Google Chrome / Edge Browser

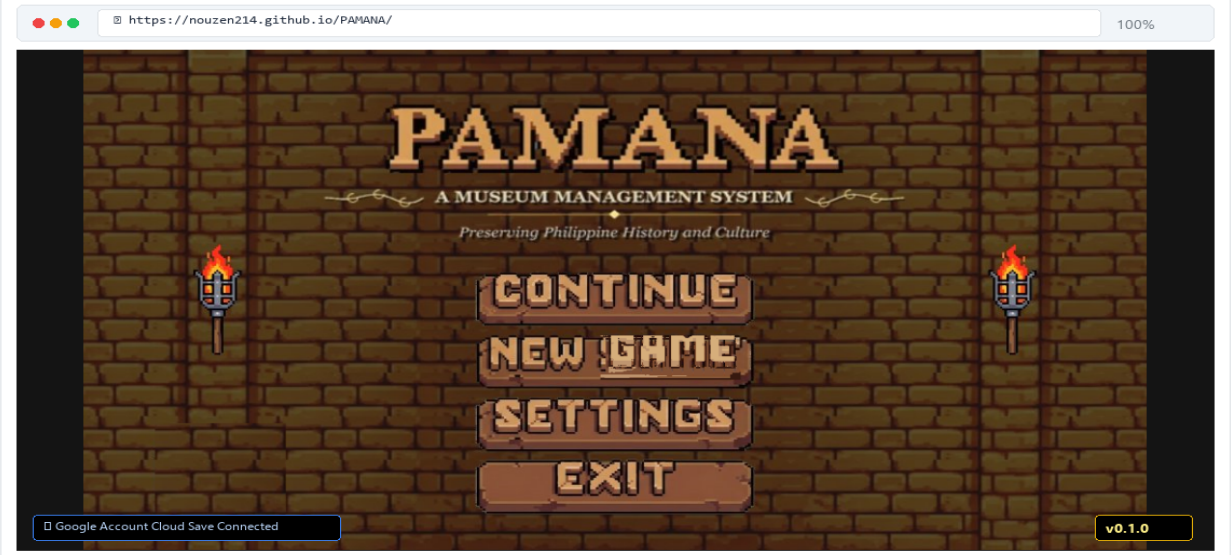
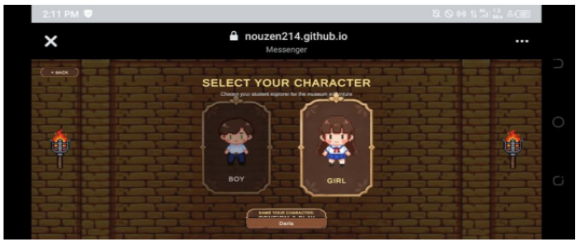


Figure 3: Live Production Deployment on GitHub Pages (https://nouzen214.github.io/PAMANA/) showing v0.1.0 identifier and Main Menu.

EVIDENCE 4: Release Readiness & Critical Bug Fixing Verification

Testing Scope: Dual Character Select, Mobile Controls Ergonomics, Proximity Plaque & Artifact Mount

Dual Character Select System (Darrel & Darla)



Mobile Controls Ergonomics & Proximity 'E' Button



Release Readiness Verification Checklist & Resolved Defects

Defect / Release Requirement	Resolution & Implementation Details	Status
Lobby Portal Barrier Collision	Added solid doorway collider (LockedPortalDoor.cs) preventing room clipping.	VERIFIED PASS
Battle HUD Immediate Rendering	Restructured battle sequence so monster portrait stays hidden until challenge.	VERIFIED PASS
Mobile Touch Controls Ergonomics	Shifted joystick & bag icons to screen edges; added dedicated 'E' interaction button.	VERIFIED PASS
Field Guide Book Readability	Expanded parchment canvas and increased typography size by 40%+ for mobile readability.	VERIFIED PASS
Firebase Cloud Save Persistence	Restored Google Account binding with automated PlayerPrefs fallback mechanism.	VERIFIED PASS
Main Menu Button Hover States	Implemented pixel-art gold glowing hover transitions and dynamic Google logout button.	VERIFIED PASS
Version Display Identification	Dynamic v0.1.0 label anchored in lower-right corner and verified across all platforms.	VERIFIED PASS

Figure 4: Release Readiness & In-Game Bug Fixing Verification (CharacterSelect UI, mobile controls ergonomics, and defect resolutions).

Briefly answer:

What did you learn about the difference between developing software and releasing software?

Limit the answer to approximately 100 words.

Developing software focuses on writing code, implementing gameplay mechanics, and testing features within an ideal development environment like the Unity Editor. In contrast, releasing software demands ensuring that the application runs reliably and predictably in the hands of actual end-users. Releasing requires rigorous version control, clean asset packaging, cross-browser compatibility testing, and eliminating environment-specific dependencies. While development tolerates debug scripts and placeholders, releasing requires strict traceability, automated build verification, deployment infrastructure, and ensuring that features like cloud persistence and responsive UI perform flawlessly across diverse client devices outside the editor.

Word Count: 90 words

Release Verification Summary	Outcome
Target Platform Build Execution	**PASS (32 scenes compiled to WebGL)**
Standalone Distribution Package (ZIP + Launcher)	**PASS (Pamana-v0.1.0.zip + run_webgl.bat)**
Live Cloud Deployment & Hosting	**PASS (GitHub Pages SSL Deployed)**
Version Traceability & Git Tagging	**PASS (Tag v0.1.0 linked to commit)**
Release Readiness & Defect Resolution	**PASS (All 7 critical defects resolved)**